DepNet - EEG+Speech Multimodal Architecture FOR depression detection using MODMA

MODMA-

Modma is Multimodal open dataset for mental disorder analysis
It is a dataset containing EEG and speech recordings of clinically proven depressed as well as healthy patients

EEG

- **full brain 128-electrodes EEG experiment**: 53 participants include a total of 24 outpatients (13 males and 11 females; 16–56-year-old) diagnosed with depression, as well as 29 healthy controls (20 males and 9 females; 18–55- year-old) were recruited

- **pervasive 3-electrodes EEG experiment**: 55 participants include a total of 26 outpatients (15 males and 11 females; 16–56-year-old) diagnosed with depression, as well as 29 healthy controls (19 males and 10 females; 18– 55-year-old) were recruited

- **Audio experiment**: 52 participants include a total of 23 outpatients (16 males and 7 females; 16–56-year-old) diagnosed with depression, as well as 29 healthy controls (20 males and 9 females; 18–55- year-old) were recruited.

Full brain 128-electrodes EEG experiment is a composure of two tasks

Task 1: Resting state No experimental material.

Task 2: Dot probe The dataset we present is composed of facial pictures from the standardized native Chinese Facial Affective Picture System (CFAPS)[17]. The facial pictures were chosen and classified into four sets as fear, sad, happy, and neutral emotion based on their valence. Two different valences of facial pictures (one belonging to emotional sets, the other to neutral set) were selected arbitrarily. The stimuli pairs consisting of emotional and neutral facial pictures, appeared side by side on the screen.

pervasive 3-electrodes EEG experiment:

Only resting-state EEG was recorded; therefore, there is no experimental material used.

full brain 128-electrodes EEG experiment:

Continuous EEG signals were recorded using a 128-channel HydroCel Geodesic Sensor Net (Electrical Geodesics Inc., Oregon Eugene, USA) and Net Station acquisition software (version 4.5.4). The sampling frequency was 250 Hz

pervasive 3-electrodes EEG experiment:

Data was recorded by a 24-bit A / D converter record EEG signals at a sampling frequency of 250Hz

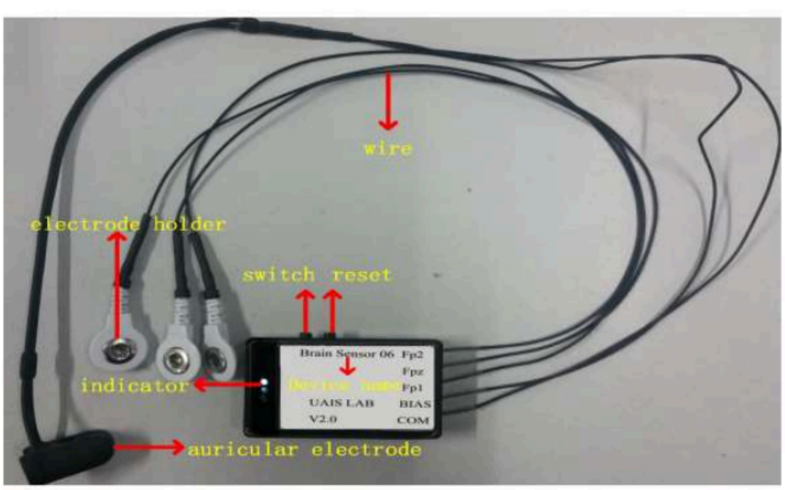


Fig 1. Three-electrode pervasive EEG collection device

Audio experiment:
The experiments were performed in a quiet, clean, soundproof, and no electromagnetic interference room. During the experiment, the ambient noise of the lab must be less than 60dB. The devices we used for recording are Neumann TLM102 (microphones) and RME FIREFACE UCX (audio card) with a 44.1 kHz sampling rate and 24-bit sampling depth. All recording data were saved as uncompressed W A V format.

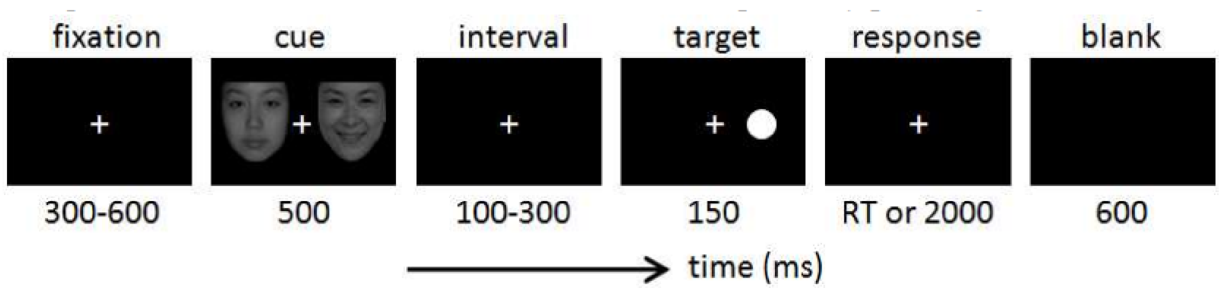
Experimental paradigm

full brain 128-electrodes EEG experiment: The multi-channel EEG was collected in a quiet, sound-proof, well-ventilated room without strong electromagnetic interference. Participants completed the tasks sitting alone in the room, while the operators were monitoring their progress in the adjoining room. When the electrode placement is completed, and the impedance meets the requirements, data acquisition can be started. All participants were asked to complete two tasks: resting state and dot-probe tasks.

Task 1: Resting state 5 minutes of eyes-closed resting-state EEG was recorded. Participants were required to keep awake and still without any bodily movements including heads or legs, as well as any unnecessary eye movements, saccades, and blinks. After completion of Task 1, participants had a rest and then completed Task 2.

Task 2: Dot probe
Participants were seated in front of the monitor (17" monitor, 1280 ×1024 resolution, and 60 Hz refresh rate) at a distance of 60 cm. All relevant instructions were shown on the computer screen initially. Before the experiment began, the participants were instructed to complete the 10 practice trials to get familiar with the task. In the formal experiment, the participants were instructed to focus their attention on the emotional-neutral face pairs with eyes viewing freely. And they were asked to press the button on the reaction box as quickly and accurately as possible when the dot appeared. The participants must press down the button without any

bodily movements including heads or legs, as well as any unnecessary eye movements, saccades, and blinks. After completing each block, they would have a rest



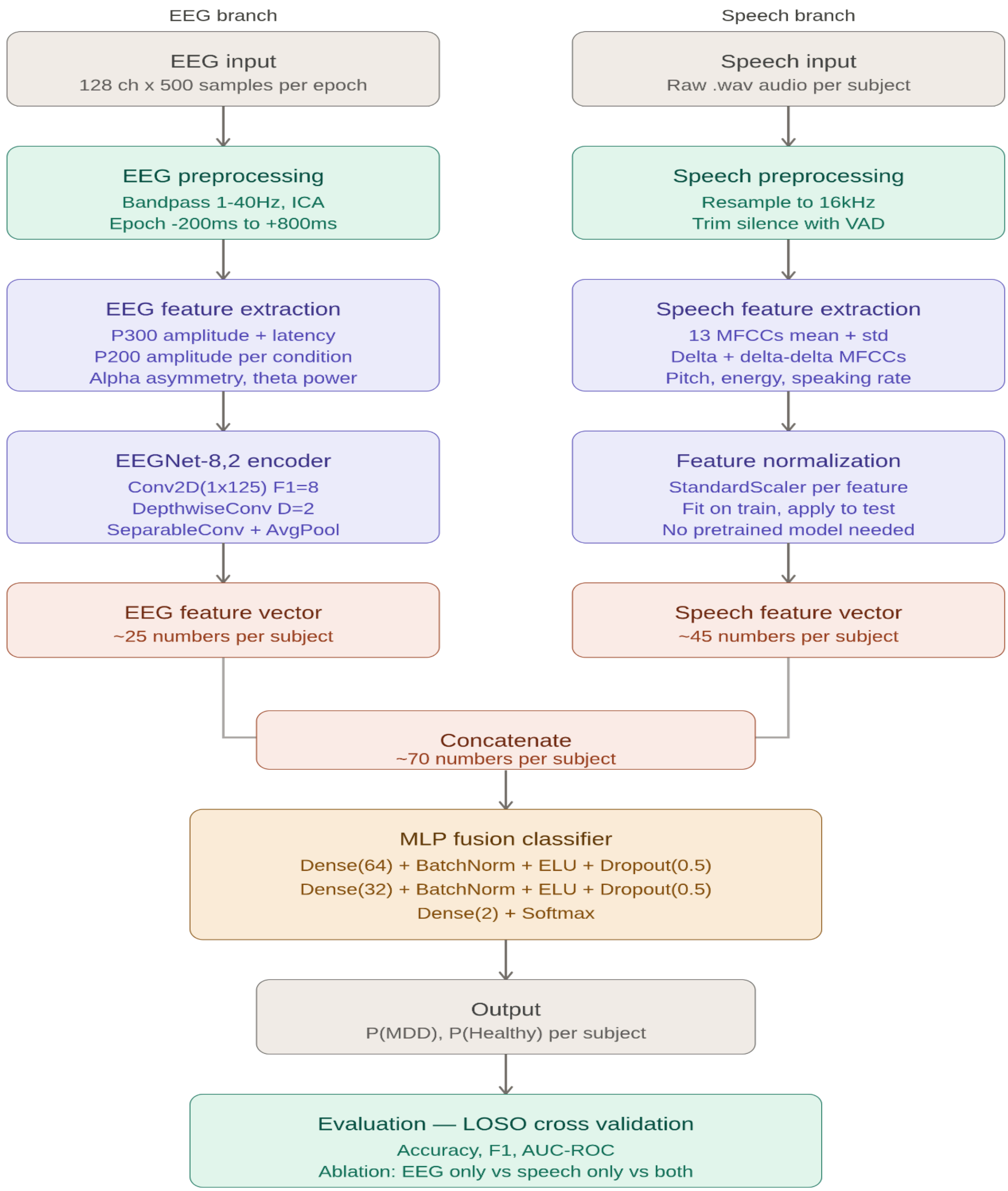
pervasive 3-electrodes EEG experiment: The data recorded in a room without loud noise and strongly magnetic. Participants kept their eyes closed until they were observed their EEG signals were relatively stable, then we started a 90-second data acquisition

Audio experiment:

There are three fixed-order parts: interview, reading, and picture description
1) Interview: This task contained 18 questions with positive, neutral and negative meanings. These topics came from DSM-IV and some depression scales which are often used in this field and often used in the acoustic analysis in international, multilingual clinical research. And three groups words with positive (e.g., outstanding, happy), neutral (e.g., center, since) and negative (e.g., depression, wail) emotion

Now the next section will contain information about the architecture used in DepNet

Architecture of DepNet



1. EEG Branch (Neurophysiological & Deep Features)

- **Preprocessing:** Continuous 128-channel data is bandpass filtered (1-40Hz), artifact-cleaned via ICA, and epoched around the Dot-probe stimulus triggers (e.g 200 ms to 800 ms).
- **Hand-Engineered ERP Extraction:** For each subject, compute the grand average ERP across target trials. Extract explicit regional peak amplitudes and latencies for the **P200** and **P300** components.
- **Deep Feature Extraction:** Train an **EEGNet-8,2** architecture on the individual trials. Once trained via Leave-One-Subject-Out (LOSO), freeze the network and pass each subject's trials through it, taking the average of the final flattened convolutional layer to obtain a stable, deep representation.
- **Output:** A unified **Subject EEG Vector** combining both hand-engineered ERP metrics and deep EEGNet embeddings.

2. Speech Branch (Acoustic & Prosodic Features)

- **Feature Extraction:** Use librosa to compute 13 MFCCs, along with their first and second derivatives over the audio segments. Extract fundamental frequency, root-mean-square energy, and speech rate metrics.
- **Statistical Aggregation:** Compute the global mean and standard deviation of these features across the entire duration of the subject's recording session.
- **Output:** A static, low-dimensional **Subject Speech Vector** that captures macro-level acoustic and prosodic indicators of depressive states without the risk of high-dimensional overfitting.

3. Intermediate Fusion & Classification Head

- **Concatenation:** For each subject, merge the Subject EEG Vector and Subject Speech Vector into a single, comprehensive multimodal feature vector.

- **Classifier:** A compact Multi-Layer Perceptron (MLP) with 2 hidden layers, utilizing aggressive Dropout (p=0.5) on the hidden states and Batch Normalization to prevent the network from memorizing subject identities.
- **Optimization: AdamW** optimizer paired with binary cross-entropy loss.